

JINGSHU PENG

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EDUCATION

Carnegie Mellon University

M.S. in Mechanical Engineering (Research)

Pittsburgh, PA

Expected Jun 2027

Tsinghua University

B.Eng. in Energy and Power Engineering (Aerospace)

Beijing, China

Jun 2025

B.S. in Theoretical and Applied Mechanics

SKILLS

Programming & Systems: C++, Python, Linux, Git, multithreading, concurrency, debugging, performance optimization

Robotics: ROS2, MoveIt, TF2, motion planning, state machines, hardware interfaces, feedback control, MuJoCo, Webots

Perception: OpenCV, camera calibration, point clouds, 3D registration, ICP, multi-view vision

Robot Learning: PyTorch, imitation learning, behavior cloning, reinforcement learning, Transformers, cross-attention

PROFESSIONAL EXPERIENCE

Instinct Robotics

Pittsburgh, PA

Robotics Software Engineer Intern

May 2026-Aug 2026

- Developed and deployed **production C++ and Python software for industrial robots in ROS2**, integrating perception, motion planning, state machines, and hardware interfaces.
- Reduced perception runtime by **~30%** by optimizing point-cloud processing and ICP registration through algorithmic improvements, multithreading, and ROS2 node-level caching.
- Improved end-to-end robot cycle time by **20%** by profiling and optimizing perception and execution pipelines across multiple ROS2 components.
- Resolved cross-stack failures across perception, planning, ROS2, and hardware through on-robot testing.

OAG Technology

New York, NY

Data Scientist Intern - Smart Bidding

May 2025-Aug 2025

- Developed production **RL bidding policies** using contextual bandits, PPO, and TD3, improving ROI by 18.5%.
- Built low-latency inference and data pipelines supporting **100K+ QPS**.

SELECTED TECHNICAL PROJECTS

Multi-View Imitation Learning for Robotic Manipulation

Pittsburgh, PA

Carnegie Mellon University

Jan 2026-May 2026

- Developed a **Transformer-based imitation-learning policy for real-world manipulation** on a **Unitree Z1 Pro 6-DoF arm**, using behavior cloning, cross-attention, and action chunking over multi-view RGB and proprioceptive observations.
- Built a synchronized **multi-view demonstration pipeline** using SpaceMouse teleoperation, IK, and multi-camera observations; trained a **~45M-parameter policy** for 200K steps with **~100 Hz real-robot inference**.
- Evaluated **cross-view generalization**, with multi-view training improving novel-view success from **0% to 42.9%**.

Real-Time Estimation Pipeline for Adaptive Exoskeleton Control

Pittsburgh, PA

Carnegie Mellon University

Aug 2025-Present

- Developed a **real-time streaming pipeline** from IMU data for online biomechanical estimation, integrating signal processing, low-latency PyTorch inference, and runtime reliability checks for hardware deployment.
- Developing an **adaptive exoskeleton control system** integrating real-time inference, gait-event detection, feedback control, and hardware execution to adjust assistance magnitude and timing online.

Dangerous Pedestrian Model for Autonomous Driving Simulation Scenarios

Beijing, China

Tsinghua University

Aug 2024-Dec 2024

- Developed an **SAC-LSTM learned control policy** from real-world trajectory data for safety-critical autonomous-driving simulation, achieving **20–30% higher realism/aggressiveness** than rule-based baselines.

Amphibious Biomimetic Robot with Combined BCF-MPF Propulsion System

Beijing, China

Tsinghua University

Aug 2022-Dec 2023

- Led a 6-member team developing an amphibious robot; implemented **embedded C++ on STM32**, integrated sensors and actuators, and performed **hardware bring-up, system integration, and closed-loop testing**.
- Secured **\$30K** in funding, resulting in a peer-reviewed publication and an invention patent.

PUBLICATIONS & PATENTS

[1] J. Peng et al., “Multi-environment robotic locomotion through integrated pectoral and caudal fin propulsion,” *Ocean Engineering*, 2025.

[2] J. Peng* et al., “Center of mass estimation during non-cyclic activities: Comparison of marker-based methods and their fusion with ground reaction forces,” *Journal of Biomechanics*, 2025.

[3] Chinese Invention Patent, “Amphibious Biomimetic Robot with Dual Propulsion System,” under substantive examination.